Unified Funeral Services Management Platform

Conceptual Layered Architecture (FSD-01 → FSD-31)

User & Channel Layer

- Web Portal (Internal)
- Mobile App (Staff)
- Family Portal
- Partner Portal
- Regulatory Interface
- Chatbot Interface
- Document Scanner
- Kiosk / Tablet
- API Gateway
- Bulk Data Exchange

AI & Intelligence Layer

- Obituary & Memorial AI FSD-08
- Handwriting Recognition AI FSD-09
- Multilingual Consultation AI FSD-10
- Predictive Analytics Engine
- Cultural / Religious Customization FSD-24
- Anomaly Detection & Data Quality FSD-16
- NLP & Semantic Knowledge Graph FSD-18
- Recommendation Engine FSD-04

Core Business Services Layer

Core Operations Domain (FSD-01 → FSD-06)

- Case Management FSD-01
- Member Registration FSD-02
- Chain of Custody & QR FSD-03
- Service & Ceremony Planning FSD-04
- Payment & Financial Mgmt FSD-05
- Inventory & Merchandise FSD-06
- Scheduling & Resource Mgmt
- Staff & Task Management
- Fleet & Equipment Telemetry FSD-26
- Vendor & Partner Mgmt

Customer Experience Domain (FSD-07 → FSD-10)

- Family Portal FSD-07
- AI Obituary & Memorial Assistant FSD-08
- Handwritten Document Scanner FSD-09
- AI Consultation Chatbot FSD-10

Data Governance & Management Layer

Data Governance Domain (FSD-11 → FSD-18)

- Success Metrics FSD-11
- RBAC FSD-12
- Data Dictionary FSD-13
- Data Lifecycle FSD-14
- Data Ingestion FSD-15
- Data Quality FSD-16
- Data Governance FSD-17
- Metadata & Ontology FSD-18
- Master Data Mgmt
- Reference Data Mgmt
- Data Lineage
- Privacy & Consent

Data Storage & Persistence

- Operational DB (OLTP)
- Data Warehouse (OLAP)
- Document Store (Blob/S3)
- Analytics Data Lake
- Audit & Log Store (Immutable)

Analytics & Insights Layer (FSD-19 → FSD-22)

- Visitor Analytics FSD-19
- Data Observability FSD-20
- Analytics Platform FSD-21
- KPI Metrics Framework FSD-22
- Executive Dashboards
- Operational Reports
- Predictive Models
- Embedded BI

Integration & Interoperability Layer (FSD-25, FSD-31)

- Third-Party Integration Hub FSD-25
- External Data Interface FSD-31
- API Management & Gateway
- Government Registries
- Insurance Carriers
- Cemeteries
- Crematories
- Accounting Platforms
- Payment Processors
- Florists
- Legacy Funeral Homes
- IoT Devices
- OBD-II Telemetry
- Partner Systems
- Regulatory Agencies

Security, Compliance & Audit Layer

Security & Audit Domain (FSD-27, FSD-28)

- Audit Trail & Compliance Logging FSD-27
- Data Encryption & Privacy FSD-28
- Identity & Access Management (IAM)
- GDPR / CCPA Compliance
- Key Management & Secrets
- Threat Detection & Response

Global Compliance Domain (FSD-23, FSD-24)

- Multi-Jurisdiction Regulatory Compliance FSD-23
- Cultural & Religious Customization FSD-24
- Regulatory Reporting Automation

CMMI Level 5 Process Excellence Layer (FSD-29, FSD-30)

Continuous Process Improvement (FSD-29)

- Causal Analysis
- Innovation Management
- Quantitative Process Mgmt

Quantitative Project Management (FSD-30)

- Statistical Process Control
- Predictive Modeling

Infrastructure & Platform Layer

- Cloud Infrastructure
- Container Orchestration
- Message Queues
- API Gateway
- CDN & Edge
- Monitoring & Logging
- CI/CD Pipeline
- Backup & Disaster Recovery
- Load Balancing
- Service Mesh